ASSIGNMENT NO 2

by

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Submitted to: Miss Haseena Kainat
Subject: Object Oriented Programming
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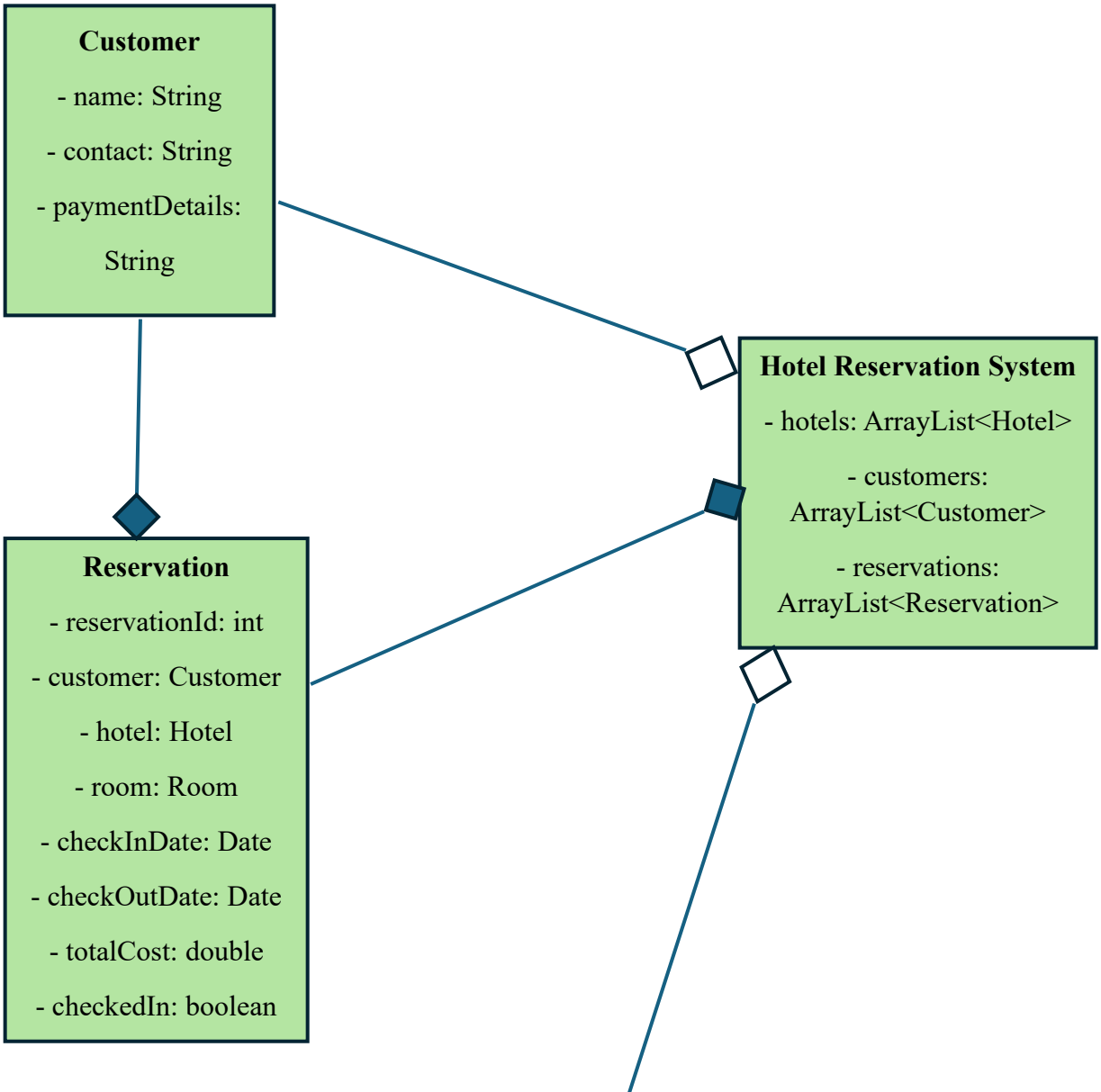
Question 1:

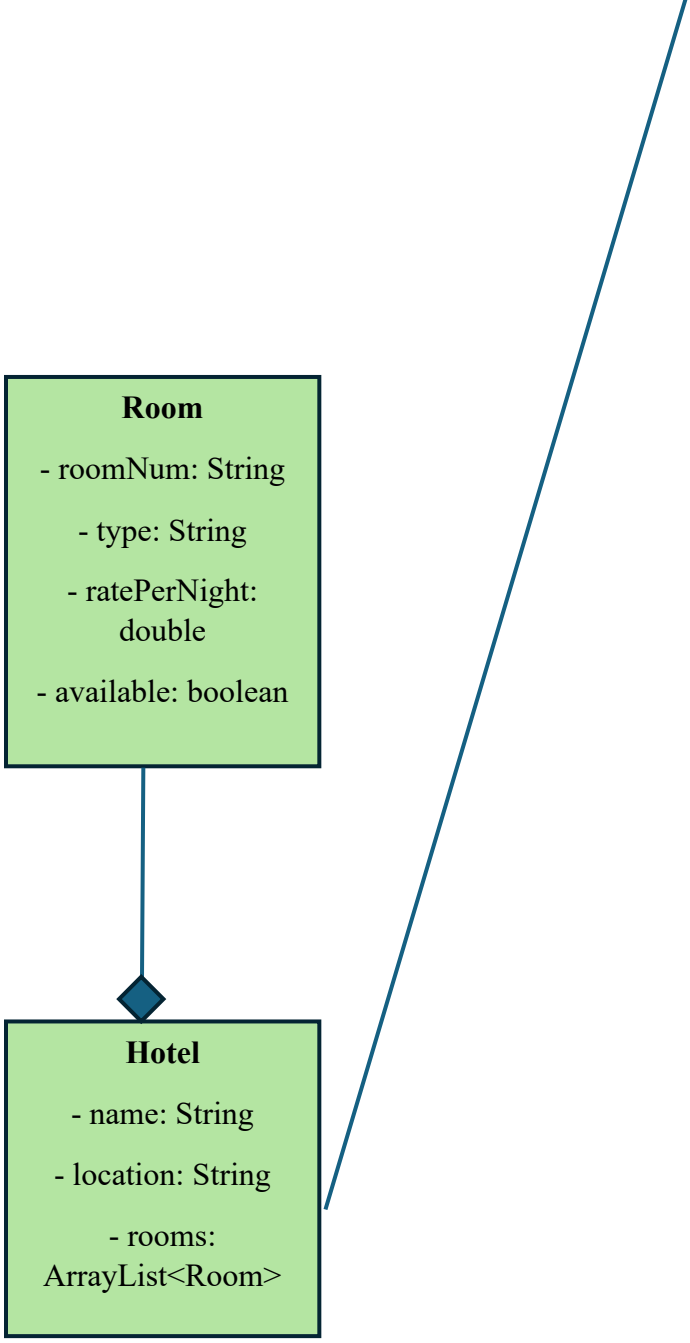
A hotel reservation system allows customers to search for available rooms, book them, and manage their reservations. The system supports multiple hotels, and each hotel offers different types of rooms such as single, double, and suite. Each room has a set of attributes like room number, type, rate per night, and availability status.

Customers must register in the system, providing personal details like name, contact information, and payment details. They can search for hotels based on location, check-in and check-out dates, and room preferences (e.g., room type or price range). Once a suitable room is found, customers can make a reservation, which includes information about the reservation date, duration of stay, and total cost. Hotels can manage room availability and update room details. They can also track reservations and mark rooms as occupied or available based on the check-in/check-out process.

Customers can cancel reservations or modify them if needed. The system also allows customers to check in online and generate an invoice at the end of their stay."

UML Diagram:




Room
- roomNum: String
- type: String
- ratePerNight: double
- available: boolean
Hotel
- name: String
- location: String
- rooms: ArrayList<Room>

Code:

```java
package oopAssign2;

import java.util.*;

class Customer {
    private String name;
    private String contact;
    private String paymentDetails;

    public Customer(String name, String contact, String paymentDetails) {
        this.name = name;
        this.contact = contact;
        this.paymentDetails = paymentDetails;
    }

    public String getName() { return name; }
    public String getContact() { return contact; }
    public String getPaymentDetails() { return paymentDetails; }
}

class Room {
    private String roomNumber;
    private String type;
    private double ratePerNight;
    private boolean available;

    public Room(String roomNumber, String type, double ratePerNight) {
        this.roomNumber = roomNumber;
        this.type = type;
        this.ratePerNight = ratePerNight;
        this.available = true;
    }

    public String getRoomNumber() { return roomNumber; }
```

```java
class Room {

    public String getRoomNumber() { return roomNumber; }
    public String getType() { return type; }
    public double getRatePerNight() { return ratePerNight; }
    public boolean isAvailable() { return available; }

    public void setAvailable(boolean available) { this.available = available; }
}

class Hotel {
    private String name;
    private String location;
    private ArrayList<Room> rooms;

    public Hotel(String name, String location) {
        this.name = name;
        this.location = location;
        this.rooms = new ArrayList<>();
    }

    public String getName() { return name; }
    public String getLocation() { return location; }
    public ArrayList<Room> getRooms() { return rooms; }

    public void addRoom(Room room) {
        rooms.add(room);
    }

    public ArrayList<Room> searchAvailableRooms(String type, double maxRate) {
        ArrayList<Room> availableRooms = new ArrayList<>();
        for (Room room : rooms) {
            if (room.isAvailable() &&
                (type == null || room.getType().equals(type)) &&
```

```java
class Hotel {
    public ArrayList<Room> searchAvailableRooms(String type, double maxRate) {
        }
        }
        return availableRooms;
    }
}

class Reservation {
    private static int nextId = 1;
    private int reservationId;
    private Customer customer;
    private Hotel hotel;
    private Room room;
    private Date checkInDate;
    private Date checkOutDate;
    private double totalCost;
    private boolean checkedIn;

    public Reservation(Customer customer, Hotel hotel, Room room, Date checkInDate, Date checkOutDate) {
        this.reservationId = nextId++;
        this.customer = customer;
        this.hotel = hotel;
        this.room = room;
        this.checkInDate = checkInDate;
        this.checkOutDate = checkOutDate;
        this.room.setAvailable(available:false);
        this.checkedIn = false;

        long days = (checkOutDate.getTime() - checkInDate.getTime()) / (1000 * 60 * 60 * 24);
        this.totalCost = days * room.getRatePerNight();
    }

    public int getReservationId() { return reservationId; }
```

```java
class Reservation {

    public int getReservationId() { return reservationId; }
    public Customer getCustomer() { return customer; }
    public Room getRoom() { return room; }
    public double getTotalCost() { return totalCost; }
    public boolean isCheckedIn() { return checkedIn; }

    public void cancel() {
        room.setAvailable(available:true);
    }

    public void checkIn() {
        checkedIn = true;
    }

    public void checkOut() {
        room.setAvailable(available:true);
        checkedIn = false;
    }

    public String generateInvoice() {
        return "Invoice for Reservation #" + reservationId +
               "\nCustomer: " + customer.getName() +
               "\nHotel: " + hotel.getName() +
               "\nRoom: " + room.getRoomNumber() +
               "\nTotal Cost: $" + totalCost;
    }
}

class HotelReservationSystem {
    private ArrayList<Hotel> hotels;
    private ArrayList<Customer> customers;
    private ArrayList<Reservation> reservations;
```

```java
}

class HotelReservationSystem {
    private ArrayList<Hotel> hotels;
    private ArrayList<Customer> customers;
    private ArrayList<Reservation> reservations;

    public HotelReservationSystem() {
        hotels = new ArrayList<>();
        customers = new ArrayList<>();
        reservations = new ArrayList<>();
    }

    public void addHotel(Hotel hotel) {
        hotels.add(hotel);
    }

    public void registerCustomer(Customer customer) {
        customers.add(customer);
    }

    public ArrayList<Hotel> searchHotels(String location) {
        ArrayList<Hotel> foundHotels = new ArrayList<>();
        for (Hotel hotel : hotels) {
            if (hotel.getLocation().equalsIgnoreCase(location)) {
                foundHotels.add(hotel);
            }
        }
        return foundHotels;
    }

    public Reservation makeReservation(Customer customer, Hotel hotel, Room room, Date checkIn, Date checkOut) {
        Reservation reservation = new Reservation(customer, hotel, room, checkIn, checkOut);
        reservations.add(reservation);
```

```java
class HotelReservationSystem {

    public Reservation findReservation(int reservationId) {
        for (Reservation reservation : reservations) {
            if (reservation.getReservationId() == reservationId) {
                return reservation;
            }
        }
        return null;
    }

    public boolean cancelReservation(int reservationId) {
        Reservation reservation = findReservation(reservationId);
        if (reservation != null) {
            reservation.cancel();
            reservations.remove(reservation);
            return true;
        }
        return false;
    }
}

public class HotelDemo {
    Run | Debug
    public static void main(String[] args) {
        HotelReservationSystem system = new HotelReservationSystem();

        Hotel hotel1 = new Hotel(name:"Grand Plaza", location:"Islamabad");
        hotel1.addRoom(new Room(roomNumber:"101", type:"Single", ratePerNight:100.0));
        hotel1.addRoom(new Room(roomNumber:"102", type:"Double", ratePerNight:150.0));
        hotel1.addRoom(new Room(roomNumber:"201", type:"Suite", ratePerNight:300.0));

        Hotel hotel2 = new Hotel(name:"Sea View", location:"Karachi");
```

```java
public class HotelDemo {
    public static void main(String[] args) {

        Hotel hotel2 = new Hotel(name:"Sea View", location:"Karachi");
        hotel2.addRoom(new Room(roomNumber:"101", type:"Single", ratePerNight:120.0));
        hotel2.addRoom(new Room(roomNumber:"102", type:"Double", ratePerNight:180.0));

        system.addHotel(hotel1);
        system.addHotel(hotel2);

        Customer customer = new Customer(name:"Ali Abdullah", contact:"Ali@email.com", paymentDetails:"credit_card_123");
        system.registerCustomer(customer);

        ArrayList<Hotel> nyHotels = system.searchHotels(location:"Islamabad");
        ArrayList<Room> availableRooms = nyHotels.get(index:0).searchAvailableRooms(type:"Double", maxRate:200.0);

        if (!availableRooms.isEmpty()) {
            Date checkIn = new Date();
            Date checkOut = new Date(checkIn.getTime() + (1000 * 60 * 60 * 24 * 3));

            Reservation reservation = system.makeReservation(customer, nyHotels.get(index:0), availableRooms.get(index:0), checkIn, checkOu

            reservation.checkIn();
            System.out.println(reservation.generateInvoice());

            reservation.checkOut();
        }

        system.cancelReservation(reservationId:1);
    }
}
```

Output:

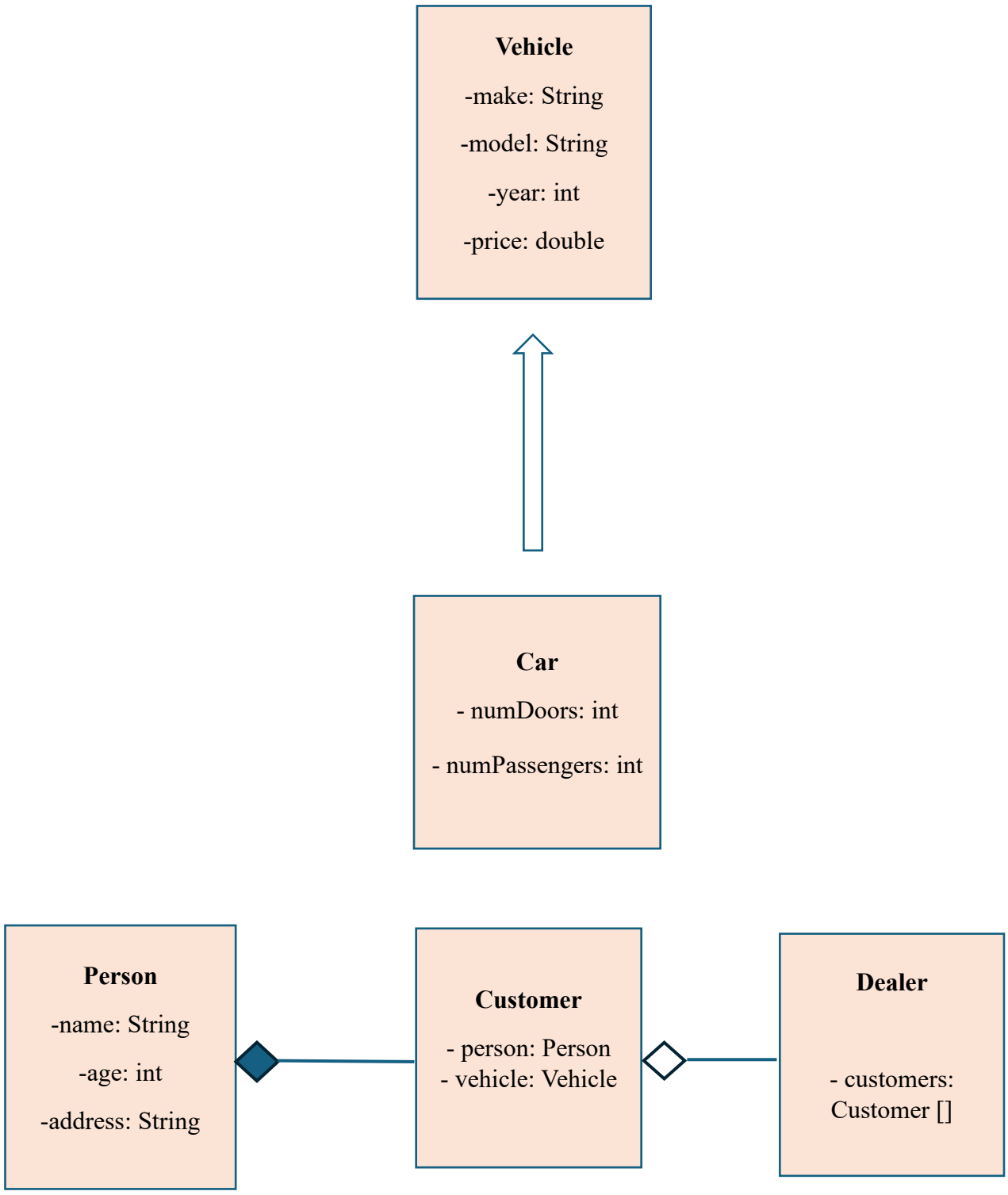
```
p\OOP\mid exam'; & 'C:\Program Files\Java\jdk-25\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Asus\A
ppData\Roaming\Code\User\workspaceStorage\77c9c9e28376ba22bd579de2f7e0219b\redhat.java\jdt_ws\mid exam_a4661313\bin' 'oopAssign2.HotelDemo'
Invoice for Reservation #1
Customer: Ali Abdullah
Hotel: Grand Plaza
Room: 102
Total Cost: $450.0
PS C:\Users\Asus\OneDrive - Higher Education Commission\Desktop\OOP\mid exam> ^C
PS C:\Users\Asus\OneDrive - Higher Education Commission\Desktop\OOP\mid exam>
PS C:\Users\Asus\OneDrive - Higher Education Commission\Desktop\OOP\mid exam>  c:; cd 'c:\Users\Asus\OneDrive - Higher Education Commission\Deskto
p\OOP\mid exam'; & 'C:\Program Files\Java\jdk-25\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Asus\A
ppData\Roaming\Code\User\workspaceStorage\77c9c9e28376ba22bd579de2f7e0219b\redhat.java\jdt_ws\mid exam_a4661313\bin' 'oopAssign2.HotelDemo'
```

Question 2:

Suppose you have a class called *Vehicle* with the following instance variables {String make, String model, int year and double price}. Now, you want to create a class called *Car* that inherits from Vehicle and has additional instance variables for numDoors and numPassengers.

In addition, you want to create a class called *Person* with the following instance variables {String name, int age, String address} You want to create another class called *Customer* that has a Person object as an instance variable, as well as a Vehicle object (either a Vehicle or a Car object). Finally, you want to create a class called *Dealer* that has an array of Customer objects as an instance variable.

UML Diagram:



Relationships:

- Car extends Vehicle (inheritance)
- Customer has-a Person (composition)
- Customer has-a Vehicle (composition)
- Dealer has-many Customer (aggregation)

Code:

```java
package oopAssign2;

class Vehicle {
    private String make;
    private String model;
    private int year;
    private double price;

    public Vehicle(String make, String model, int year, double price) {
        this.make = make;
        this.model = model;
        this.year = year;
        this.price = price;
    }

    public String getMake() { return make; }
    public void setMake(String make) { this.make = make; }

    public String getModel() { return model; }
    public void setModel(String model) { this.model = model; }

    public int getYear() { return year; }
    public void setYear(int year) { this.year = year; }

    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }
}

class Car extends Vehicle {
    private int numDoors;
    private int numPassengers;

    public Car(String make, String model, int year, double price, int numDoors, int numPassengers) {
```

File Edit Selection View Go Run ··· mid exam

Welcome | EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java

oopAssign2 > Inheritence.java > Vehicle

```java
class Car extends Vehicle {
    public Car(String make, String model, int year, double price, int numDoors, int numPassengers) {
        super(make, model, year, price);
        this.numDoors = numDoors;
        this.numPassengers = numPassengers;
    }

    public int getNumDoors() { return numDoors; }
    public void setNumDoors(int numDoors) { this.numDoors = numDoors; }

    public int getNumPassengers() { return numPassengers; }
    public void setNumPassengers(int numPassengers) { this.numPassengers = numPassengers; }
}

class Person {
    private String name;
    private int age;
    private String address;

    public Person(String name, int age, String address) {
        this.name = name;
        this.age = age;
        this.address = address;
    }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public int getAge() { return age; }
    public void setAge(int age) { this.age = age; }

    public String getAddress() { return address; }
    public void setAddress(String address) { this.address = address; }
```

Java: Ready | Ln 3, Col 1 | Spaces: 4 | UTF-8 | CRLF | Java | Go Live | Prettier

19° Search 10:41 PM 10/25/2025

File Edit Selection View Go Run ··· mid exam

Welcome | EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java

oopAssign2 > Inheritence.java > Vehicle

```java
class Person {
}

class Customer {
    private Person person;
    private Vehicle vehicle;

    public Customer(Person person, Vehicle vehicle) {
        this.person = person;
        this.vehicle = vehicle;
    }

    public Person getPerson() { return person; }
    public void setPerson(Person person) { this.person = person; }

    public Vehicle getVehicle() { return vehicle; }
    public void setVehicle(Vehicle vehicle) { this.vehicle = vehicle; }
}

class Dealer {
    private Customer[] customers;

    public Dealer(Customer[] customers) {
        this.customers = customers;
    }

    public Customer[] getCustomers() { return customers; }
    public void setCustomers(Customer[] customers) { this.customers = customers; }
}
```

Java: Ready | Ln 3, Col 1 | Spaces: 4 | UTF-8 | CRLF | Java | Go Live | Prettier

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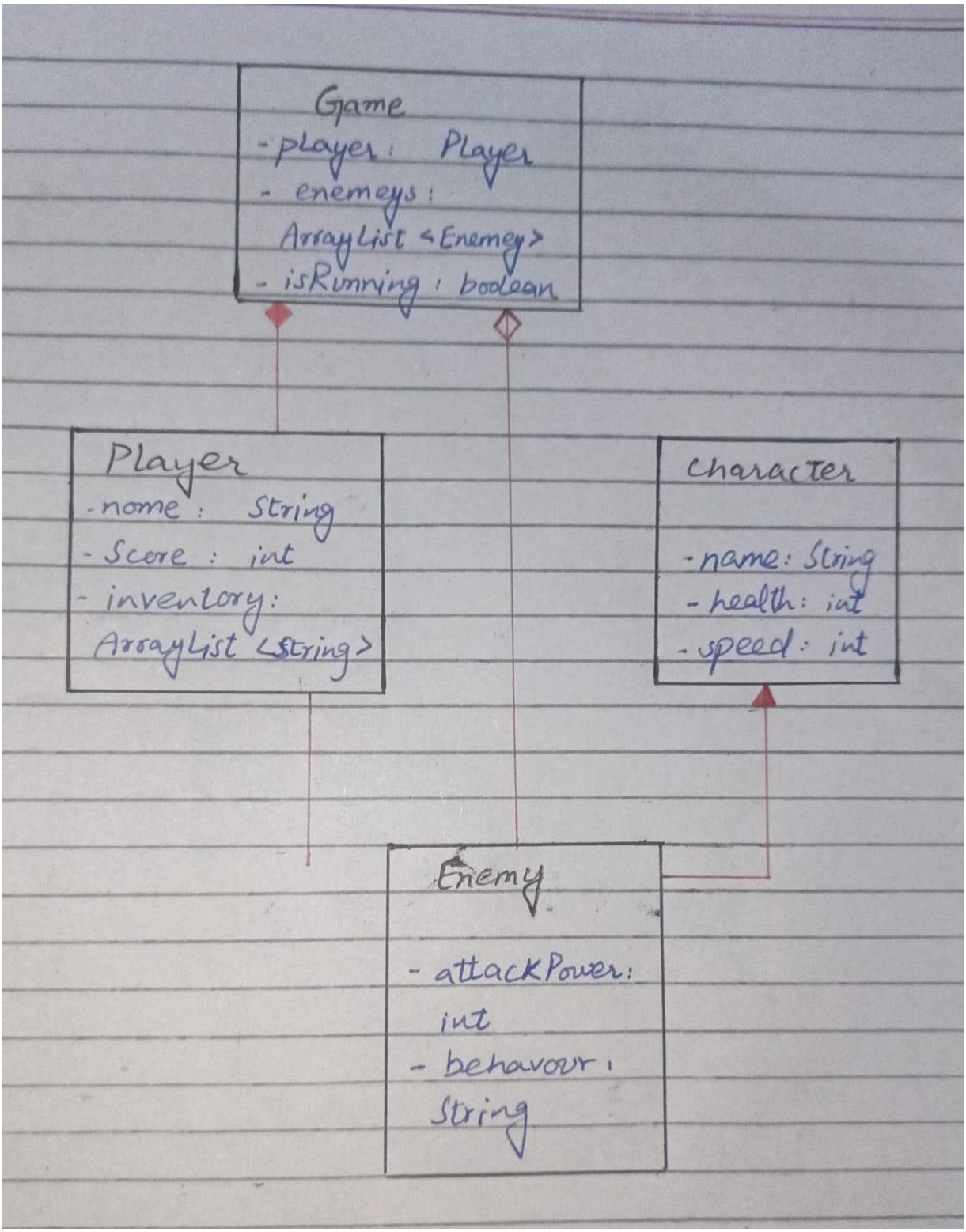
Question 3:

Suppose you are designing a simple game engine in Java. Define a class hierarchy for your game engine that includes the following classes: Game, Player, Character, and Enemy.

The Game class should contain the main game loop, which updates the game state and renders the game screen. The Player class should contain the player's name, score, and inventory. The Character class should contain the common attributes and behaviors for all characters in the game, such as health points and movement speed. The Enemy class should inherit from the Character class and include additional attributes and behaviors specific to enemies, such as attack power and behavior patterns.

Implement a constructor for each class and define appropriate getters and setters for their attributes. Write a method in the Game class that initializes a new game by creating a player object and a set of enemy objects. Finally, write a method in the Game class that updates the game state by moving the player and enemies, checking for collisions, and updating the player's score and inventory.

UML Diagram:


Game
- player : Player
- enemeys :
ArrayList <Enemey>
- isRunning : boolean
Player
- name : String
- Score : int
- inventory :
ArrayList <String>
Character
- name : String
- health : int
- speed : int
Enemy
- attackPower :
int
- behavour :
String

Code:

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ome | EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java | Player.java

oopAssign2 > Player.java > Enemy > behavior

```java
package oopAssign2;

import java.util.*;

class Game {
    private Player player;
    private ArrayList<Enemy> enemies;
    private boolean isRunning;

    public Game() {
        this.enemies = new ArrayList<>();
        this.isRunning = false;
    }

    public Player getPlayer() { return player; }
    public void setPlayer(Player player) { this.player = player; }
    public ArrayList<Enemy> getEnemies() { return enemies; }
    public boolean isRunning() { return isRunning; }
    public void setRunning(boolean running) { isRunning = running; }

    public void initializeGame() {
        player = new Player(name:"Player1", score:0);
        enemies.add(new Enemy(name:"Goblin", health:50, speed:5, attackPower:10));
        enemies.add(new Enemy(name:"Orc", health:80, speed:3, attackPower:15));
        enemies.add(new Enemy(name:"Dragon", health:150, speed:2, attackPower:25));
        isRunning = true;
    }

    public void updateGameState() {
        if (!isRunning) return;

        for (Enemy enemy : enemies) {
            enemy.move();
```

0 ⚠ 2 Java: Ready Ln 108, Col 29 Spaces: 4 UTF-8 CRLF {} Java Go Live Prettier

18° Search 11:32 PM 10/25/2025

File Edit Selection View Go Run ··· mid exam

ome | EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java | Player.java

oopAssign2 > Player.java > Enemy > behavior

```java
class Game {
    public void updateGameState() {
        }

        for (Enemy enemy : enemies) {
            if (checkCollision(player, enemy)) {
                player.setScore(player.getScore() + 10);
                player.addToInventory("Defeated " + enemy.getName());
            }
        }
    }

    private boolean checkCollision(Player player, Enemy enemy) {
        return Math.random() > 0.7;
    }

    public void gameLoop() {
        initializeGame();
        while (isRunning) {
            updateGameState();
            try {
                Thread.sleep(millis:100);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

class Player {
    private String name;
    private int score;
    private ArrayList<String> inventory;
```

0 ⚠ 2 Java: Ready Ln 108, Col 29 Spaces: 4 UTF-8 CRLF {} Java Go Live Prettier

18° Search 11:33 PM 10/25/2025

File Edit Selection View Go Run ··· mid exam

EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java | Player.java

oopAssign2 > Player.java > ...

```java
class Player {
    private ArrayList<String> inventory;

    public Player(String name, int score) {
        this.name = name;
        this.score = score;
        this.inventory = new ArrayList<>();
    }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public int getScore() { return score; }
    public void setScore(int score) { this.score = score; }
    public ArrayList<String> getInventory() { return inventory; }

    public void addToInventory(String item) {
        inventory.add(item);
    }
}

class Character {
    private String name;
    private int health;
    private int speed;

    public Character(String name, int health, int speed) {
        this.name = name;
        this.health = health;
        this.speed = speed;
    }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
```

Ln 82, Col 1 Spaces: 4 UTF-8 CRLF {} Java Go Live Prettier

Java: Ready

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File Edit Selection View Go Run ··· mid exam

EnrollmentManagementSystem.java | SmartHomeDemo.java | ParkingSoftware.java | HotelDemo.java 2 | Inheritence.java | Player.java

oopAssign2 > Player.java > ...

```java
class Character {
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public int getHealth() { return health; }
    public void setHealth(int health) { this.health = health; }
    public int getSpeed() { return speed; }
    public void setSpeed(int speed) { this.speed = speed; }

    public void move() {
        System.out.println(name + " is moving with speed " + speed);
    }
}

class Enemy extends Character {
    private int attackPower;
    private String behavior;

    public Enemy(String name, int health, int speed, int attackPower) {
        super(name, health, speed);
        this.attackPower = attackPower;
        this.behavior = "Aggressive";
    }

    public int getAttackPower() { return attackPower; }
    public void setAttackPower(int attackPower) { this.attackPower = attackPower; }
    public String getBehavior() { return behavior; }
    public void setBehavior(String behavior) { this.behavior = behavior; }

    public void attack() {
        System.out.println(getName() + " attacks with power " + attackPower);
    }
}
```

Ln 82, Col 1 Spaces: 4 UTF-8 CRLF {} Java Go Live Prettier

Java: Ready

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